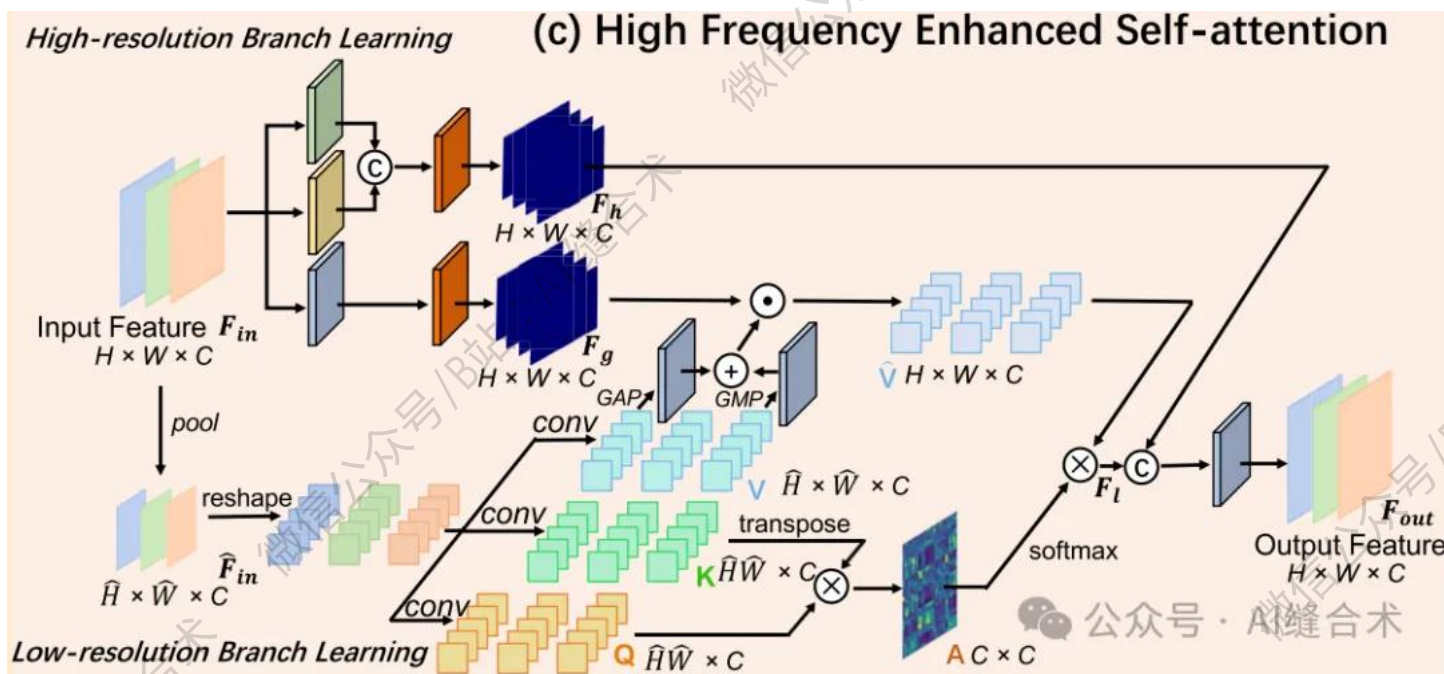


核心速览



论文题目: Ultra-High-Definition Image Restoration via High-Frequency Enhanced Transformer

中文题目: 通过高频增强 Transformer 实现超高清图像恢复

论文链接:

<https://ieeexplore.ieee.org/stamp/stamp.jsp?arnumber=11263975>

所属单位: 国防科技大学等

核心速览:

提出了一种名为 HiFormer 的双分支 Transformer 架构, 通过原生分辨率高频保留与高效上下文建模的协同, 解决超高清 (UHD) 图像恢复中 Transformer 在高频细节维护方面的挑战。

https://mp.weixin.qq.com/s/N0_LKrNf3Qp3T7q29IKwAQ

```
HFESA(  
  (dwconv_w): Conv2d(64, 64, kernel_size=(1, 1), stride=(1, 1), padding=(0, 5), groups=64)  
  (dwconv_h): Conv2d(64, 64, kernel_size=(11, 1), stride=(1, 1), padding=(5, 0), groups=64)  
  (dwconv_hw): Conv2d(128, 128, kernel_size=(3, 3), stride=(1, 1), padding=(1, 1), groups=128)  
  (conv1): Conv2d(128, 64, kernel_size=(1, 1), stride=(1, 1), bias=False)  
  (sr): AvgPool2d(kernel_size=8, stride=8, padding=0)  
  (qkv): Conv2d(64, 192, kernel_size=(1, 1), stride=(1, 1), bias=False)  
  (qkv_dwconv): Conv2d(192, 192, kernel_size=(3, 3), stride=(1, 1), padding=(1, 1), groups=192, bias=False)  
  (avg): AdaptiveAvgPool2d(output_size=1)  
  (max): AdaptiveMaxPool2d(output_size=1)  
  (conv2): Conv2d(64, 64, kernel_size=(1, 1), stride=(1, 1), bias=False)  
  (conv3): Conv2d(64, 64, kernel_size=(1, 1), stride=(1, 1), bias=False)  
  (conv4): Sequential(  
    (0): Conv2d(64, 64, kernel_size=(1, 1), stride=(1, 1), bias=False)  
    (1): Conv2d(64, 64, kernel_size=(3, 3), stride=(1, 1), padding=(1, 1), groups=64, bias=False)  
  )  
  (project_out): Conv2d(128, 64, kernel_size=(1, 1), stride=(1, 1), bias=False)  
)  
input_tensor_shape : torch.Size([2, 64, 32, 32])  
output_tensor_shape : torch.Size([2, 64, 32, 32])
```

哔哩哔哩/微信公众号：AI缝合术，独家整理！